

Graphing Quadratic Expressions (Part II)

Standard Form	Intercept Form	Vertex Form
$y = ax^2 + bx + c$	$y = a(x-m)(x-n)$	$y = a(x-h)^2 + k$

Finding the Vertex in Standard Form

If a quadratic expression is written in the form $y = ax^2 + bx + c$, then the x-value of the vertex is $-\frac{b}{2a}$.

Finding the Vertex in Intercept Form

If a quadratic expression is written in the form $y = a(x-m)(x-n)$, where m and n are real numbers, then the x-value of the vertex is $\frac{m+n}{2}$.

* Intercept form cannot be used to graph a quadratic expression if the corresponding graph does not move through the x-axis.

Finding the Vertex in Vertex Form

If a quadratic expression is written in the form $y = a(x-h)^2 + k$, then (h,k) is the vertex.

Graph the following quadratic expressions.

① $y = 2x^2 + 7x - 15$

$a = 2, b = 7, c = -15$

I) $V_x = -\frac{b}{2a}$

$V_x = -\frac{7}{2(2)}$

$V_x = -\frac{7}{4} = -1\frac{3}{4}$

$V_y = 2(-\frac{7}{4})^2 + 7(-\frac{7}{4}) - 15$

$V_y = 2(\frac{49}{16}) - \frac{49}{4} - 15$

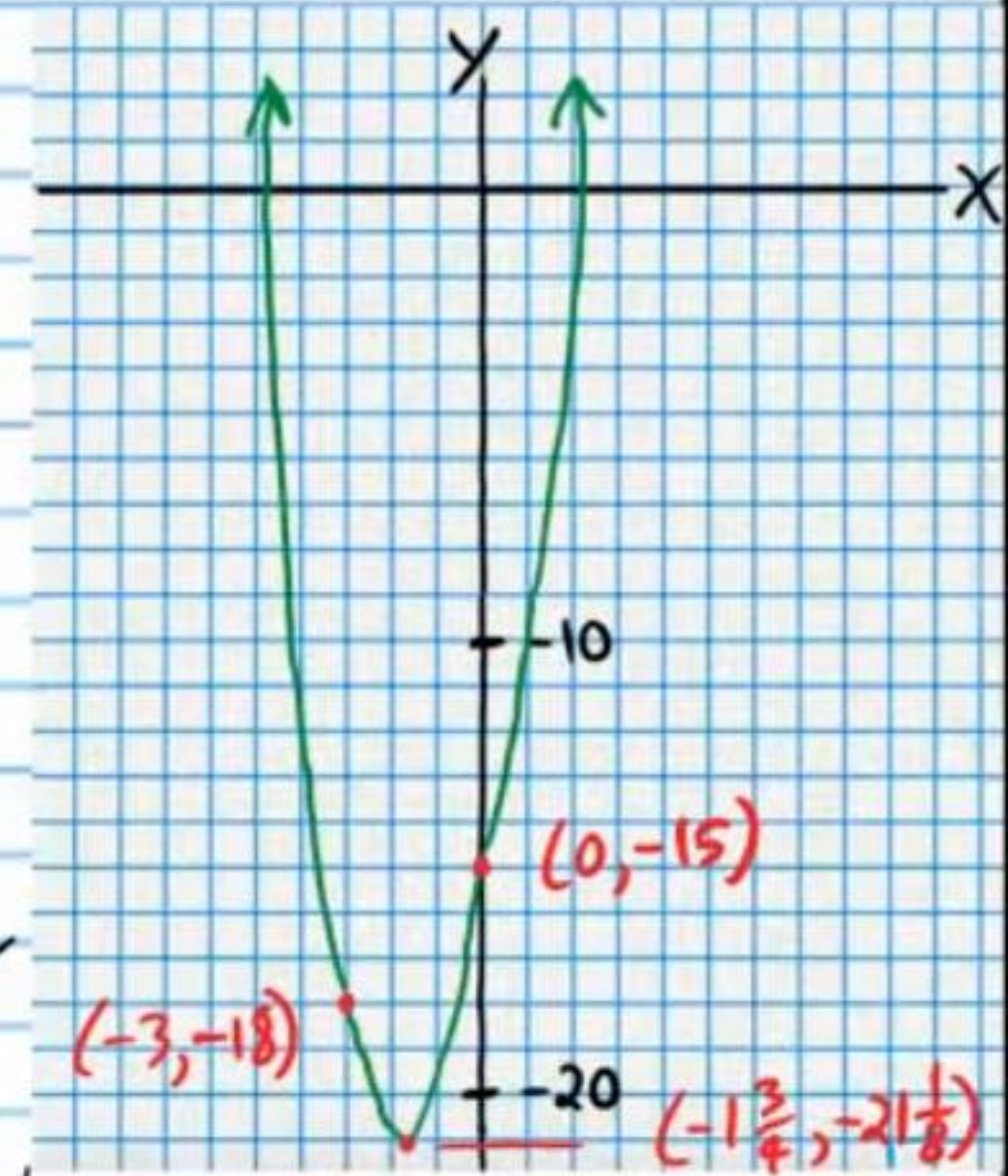
$V_y = \frac{49}{8} - \frac{49}{4} - 15$

$V_y = \frac{49}{8} - \frac{98}{8} - 15$

$V_y = -\frac{49}{8} - 15$

$V_y = -6\frac{1}{8} - 15$

$V_y = -21\frac{1}{8}$



II) $x = 0$

$y = 2(0)^2 + 7(0) - 15$

$y = -15$

III) $x = -3$

$y = 2(-3)^2 + 7(-3) - 15$

$y = 2(9) - 21 - 15$

$y = -18$

$$\textcircled{2} -\frac{1}{3}x^2 - 2x - 3 = y$$

$$a = -\frac{1}{3}, b = -2, c = -3$$

$$\text{I) } V_x = -\frac{b}{2a}$$

$$V_x = -\frac{-2}{2(-\frac{1}{3})}$$

$$V_x = \frac{1}{-\frac{1}{3}}$$

$$V_x = 1 \div (-\frac{1}{3})$$

$$V_x = 1 \cdot (-\frac{3}{1})$$

$$V_x = -3$$

$$\text{II) } x = 0$$

$$y = -\frac{1}{3}(0)^2 - 2(0) - 3$$

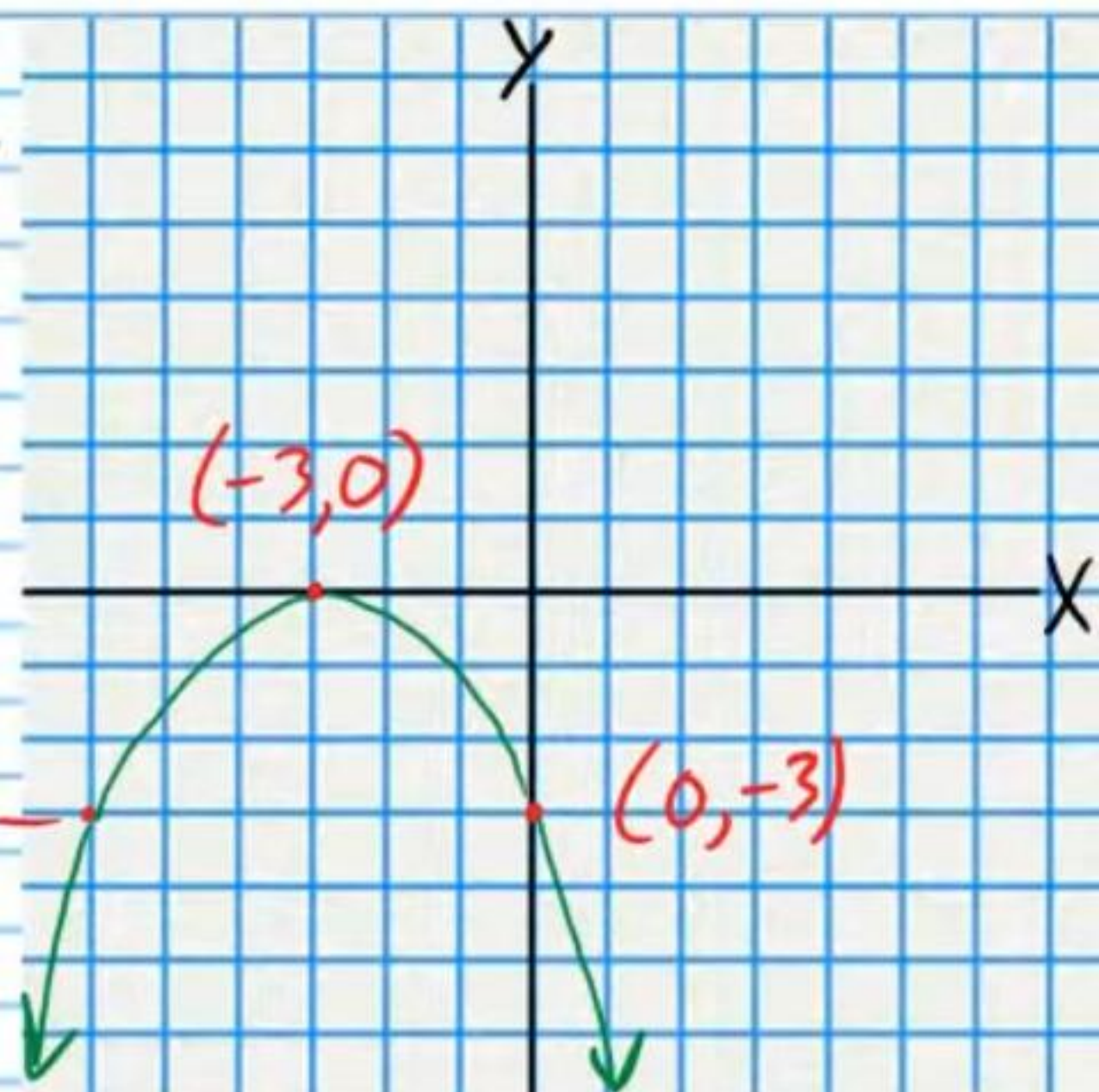
$$y = -3$$

$$V_y = -\frac{1}{3}(-3)^2 - 2(-3) - 3$$

$$V_y = -\frac{1}{3}(9) + 6 - 3$$

$$V_y = -3 + 6 - 3$$

$$V_y = 0$$



$$\textcircled{3} y = -5x^2 + 12x - 4$$

$$a = -5, b = 12, c = -4$$

$$\text{I) } V_x = -\frac{b}{2a}$$

$$V_x = -\frac{12}{2(-5)}$$

$$V_x = \frac{6}{5} = 1\frac{1}{5}$$

$$V_y = -5(\frac{6}{5})^2 + 12(\frac{6}{5}) - 4$$

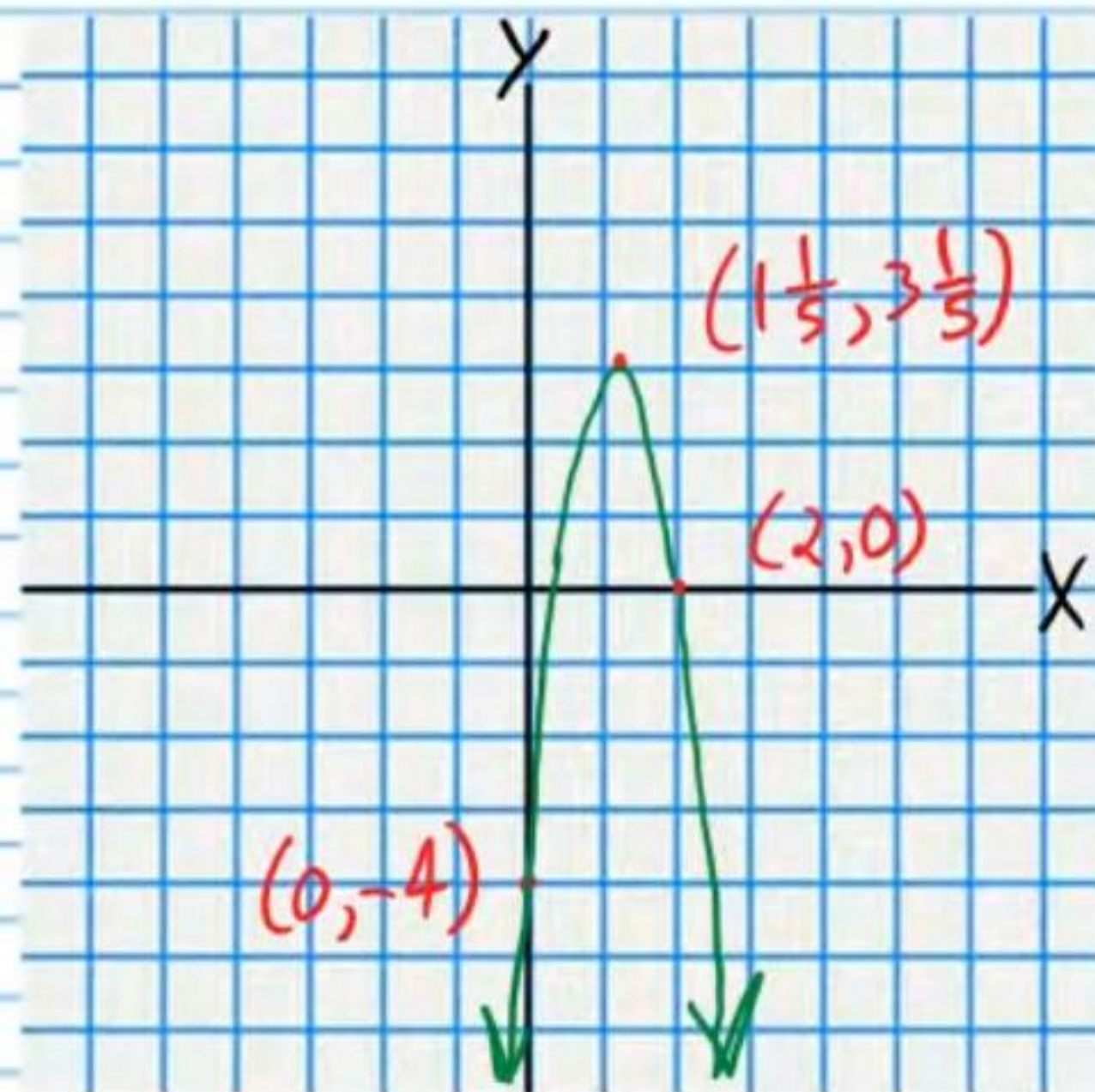
$$V_y = -5(\frac{36}{25}) + \frac{72}{5} - 4$$

$$V_y = -\frac{36}{5} + \frac{72}{5} - 4$$

$$V_y = \frac{36}{5} - 4$$

$$V_y = 7\frac{1}{5} - 4$$

$$V_y = 3\frac{1}{5}$$



$$\text{II) } x = 0$$

$$y = -4$$

$$\text{III) } x = 2$$

$$y = -5(2)^2 + 12(2) - 4$$

$$y = 0$$

$$\textcircled{4} x^2 + \frac{9}{2}x + 5 = y$$

$$a=1, b=\frac{9}{2}, c=5$$

$$\text{I) } V_x = -\frac{b}{2a}$$

$$V_x = -\frac{\frac{9}{2}}{2(1)}$$

$$V_x = -\frac{\frac{9}{2}}{2}$$

$$V_x = -\frac{9}{2} \div 2$$

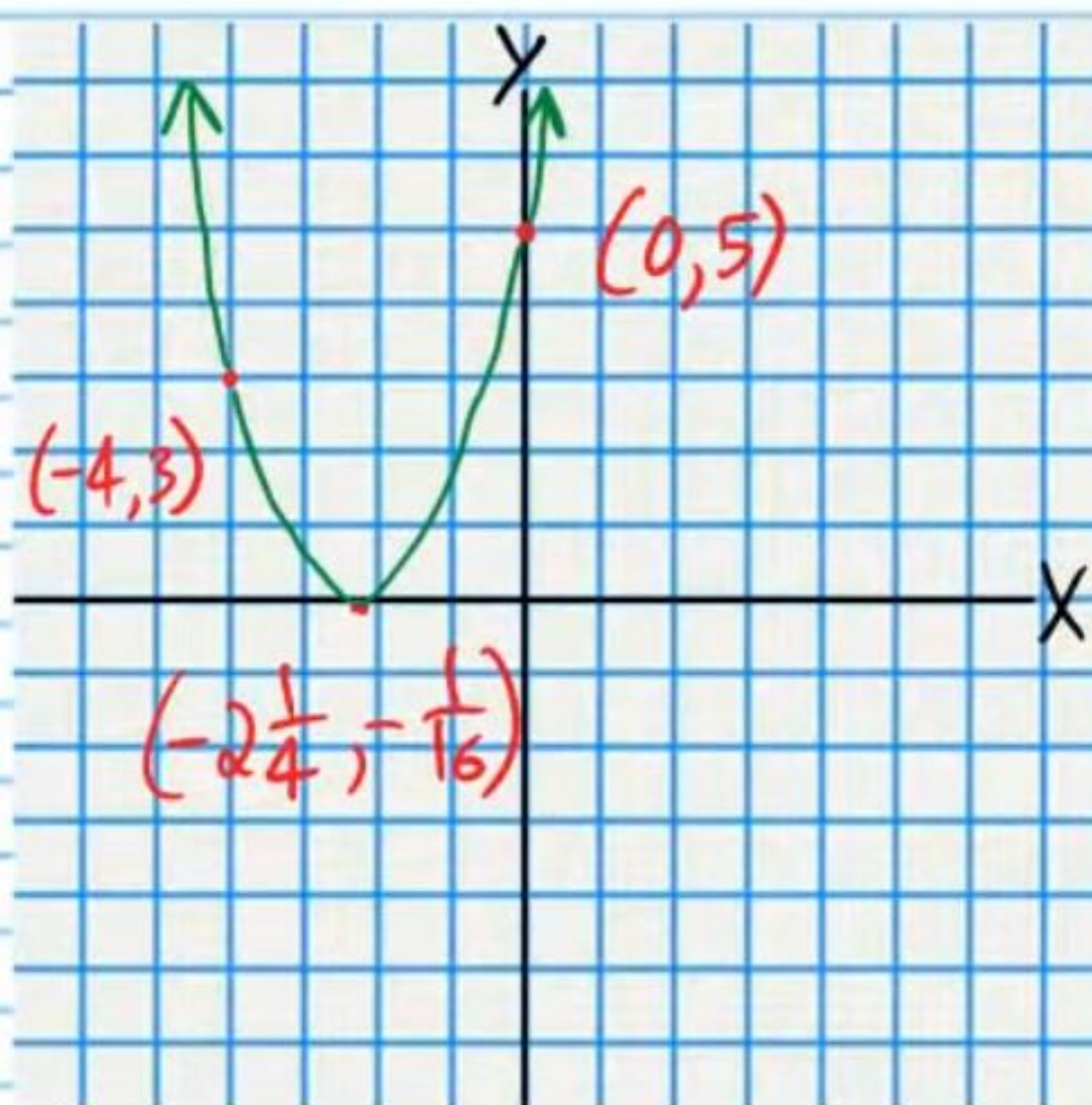
$$V_x = -\frac{9}{2} \cdot \frac{1}{2}$$

$$V_x = -\frac{9}{4} = -2\frac{1}{4}$$

$$\textcircled{V_y} = \left(-\frac{9}{4}\right)^2 + \frac{9}{2}\left(-\frac{9}{4}\right) + 5$$

$$V_y = \frac{81}{16} - \frac{81}{8} + 5$$

$$V_y = \frac{81}{16} - \frac{162}{16} + 5 = -\frac{81}{16} + 5 = -5\frac{1}{16} + 5 = -\frac{1}{16}$$



$$\text{II) } x=0$$

$$y=5$$

$$\text{III) } x=-4$$

$$y = (-4)^2 + \frac{9}{2}(-4) + 5$$

$$y = 16 - 18 + 5$$

$$y = 3$$

$$\textcircled{5} y = \frac{2}{3}(x+4)(x+2)$$

$$a=\frac{2}{3}, m=-4, n=-2$$

$$\text{I) } (-4, 0) \text{ \& } (-2, 0)$$

$$\text{II) } V_x = \frac{m+n}{2}$$

$$V_x = \frac{-4+(-2)}{2}$$

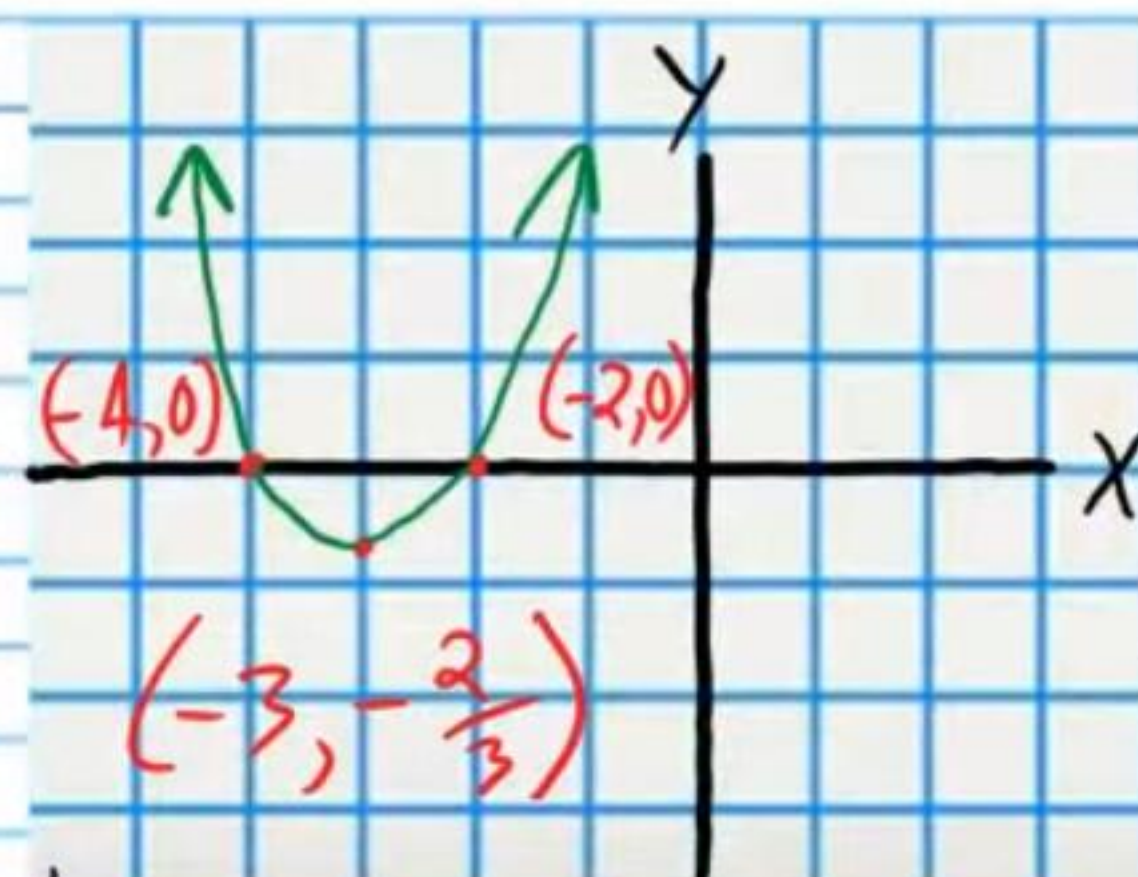
$$V_x = -\frac{6}{2}$$

$$V_x = -3$$

$$\textcircled{V_y} = \frac{2}{3}(-3+4)(-3+2)$$

$$V_y = \frac{2}{3}(1)(-1)$$

$$V_y = -\frac{2}{3}$$



$$\textcircled{6} -\frac{4}{5}(x-1)(x+5)=y$$

$$a=-\frac{4}{5}, m=1, n=-5$$

$$\text{I) } (1,0) \text{ \& } (-5,0)$$

$$\text{II) } V_x = \frac{m+n}{2}$$

$$V_x = \frac{1+(-5)}{2}$$

$$V_x = -\frac{4}{2}$$

$$V_x = -2$$

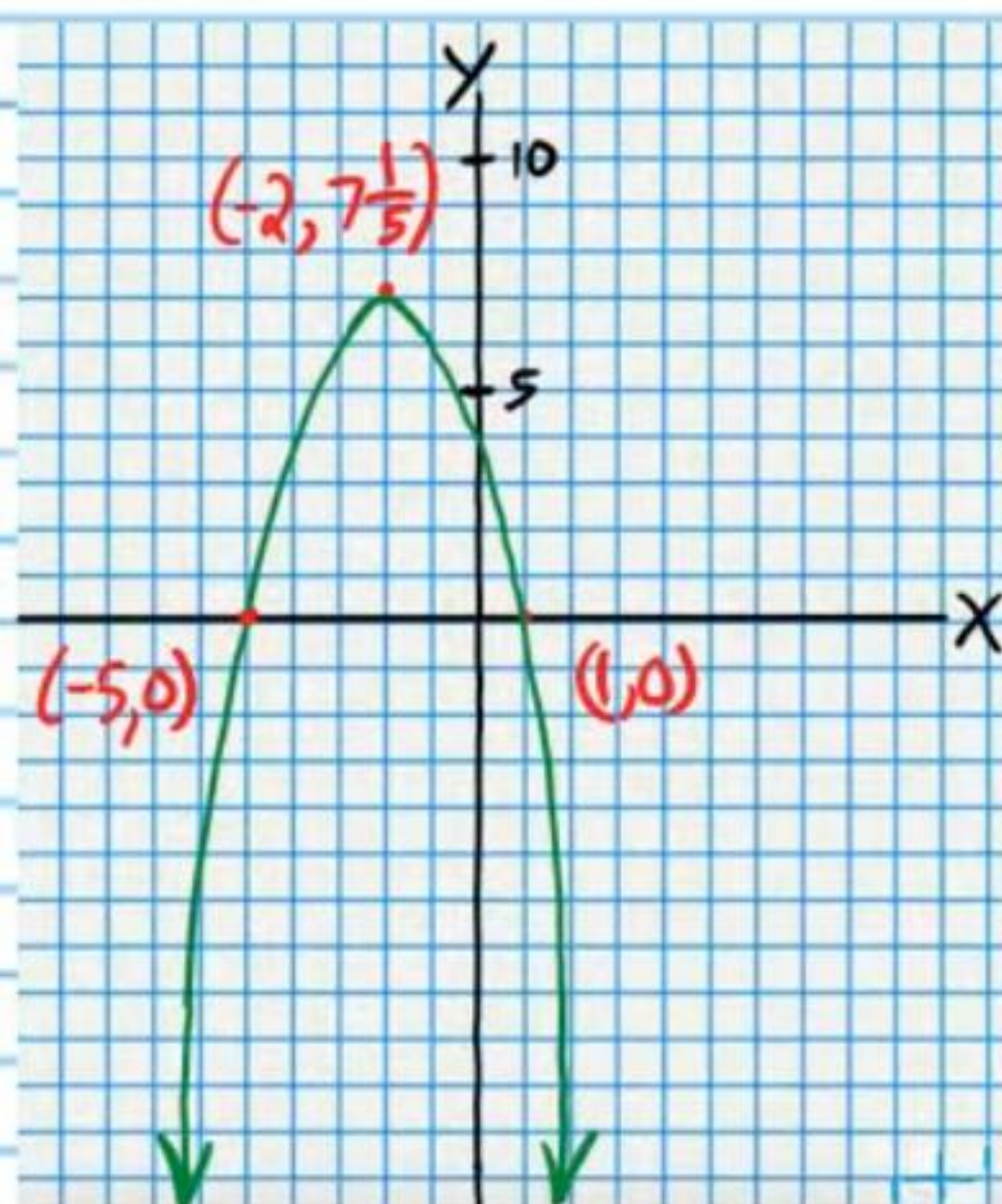
$$\text{III) } V_y = -\frac{4}{5}(-2-1)(-2+5)$$

$$V_y = -\frac{4}{5}(-3)(3)$$

$$V_y = -\frac{4}{5}(-9)$$

$$V_y = \frac{36}{5}$$

$$V_y = 7\frac{1}{5}$$



$$\textcircled{7} y = -\frac{1}{2}(x+3)^2 - 1$$

$$a=-\frac{1}{2}, h=-3, k=-1$$

$$\text{I) } (-3, -1)$$

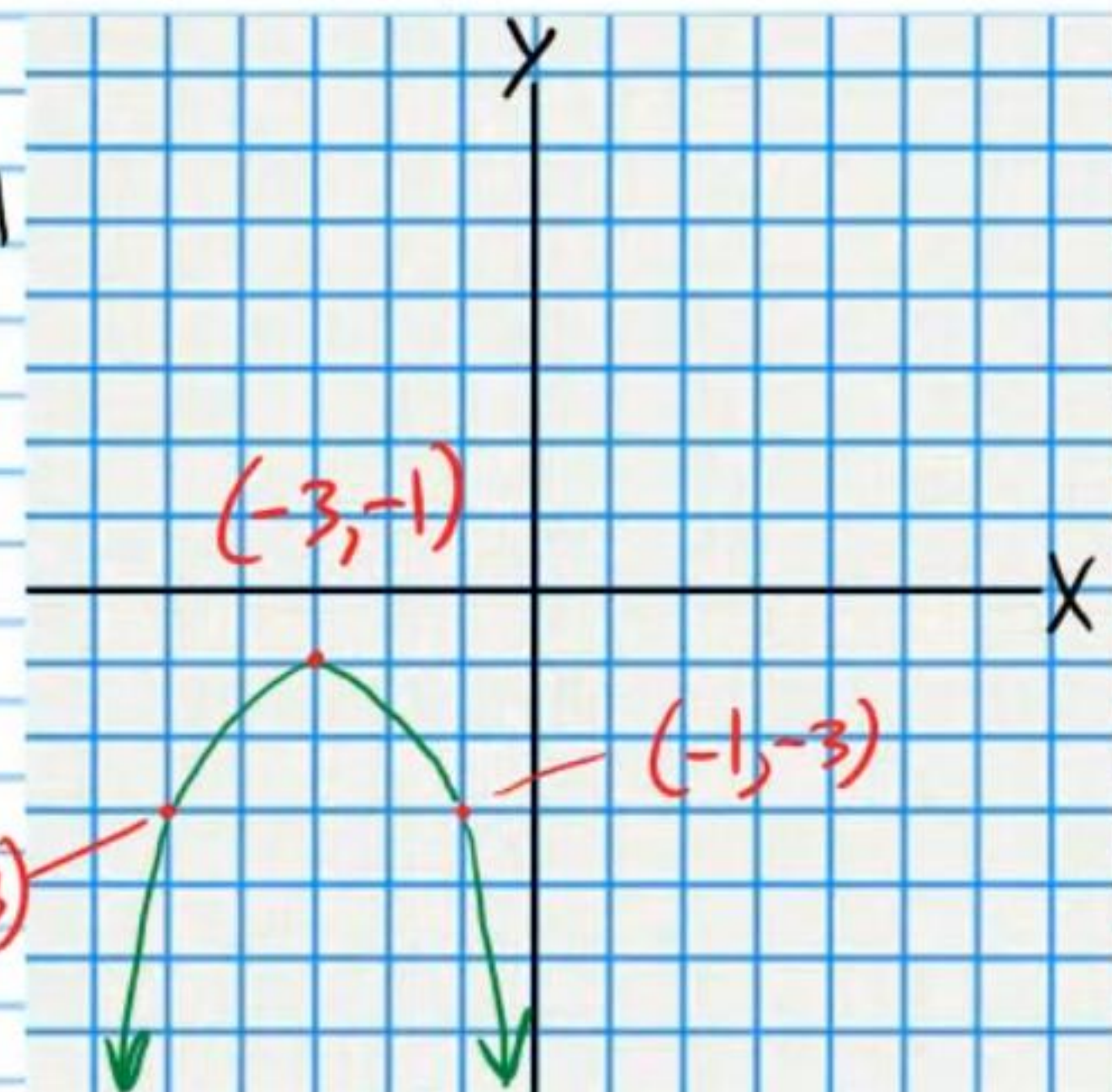
$$\text{II) } x = -1$$

$$y = -\frac{1}{2}(-1+3)^2 - 1$$

$$y = -\frac{1}{2}(2)^2 - 1$$

$$y = -\frac{1}{2}(4) - 1$$

$$y = -3$$



$$\textcircled{8} \quad \frac{2}{5}(x-3)^2 + 4 = y$$

$$a = \frac{2}{5}, h = 3, k = 4$$

$$\text{I) } (3, 4)$$

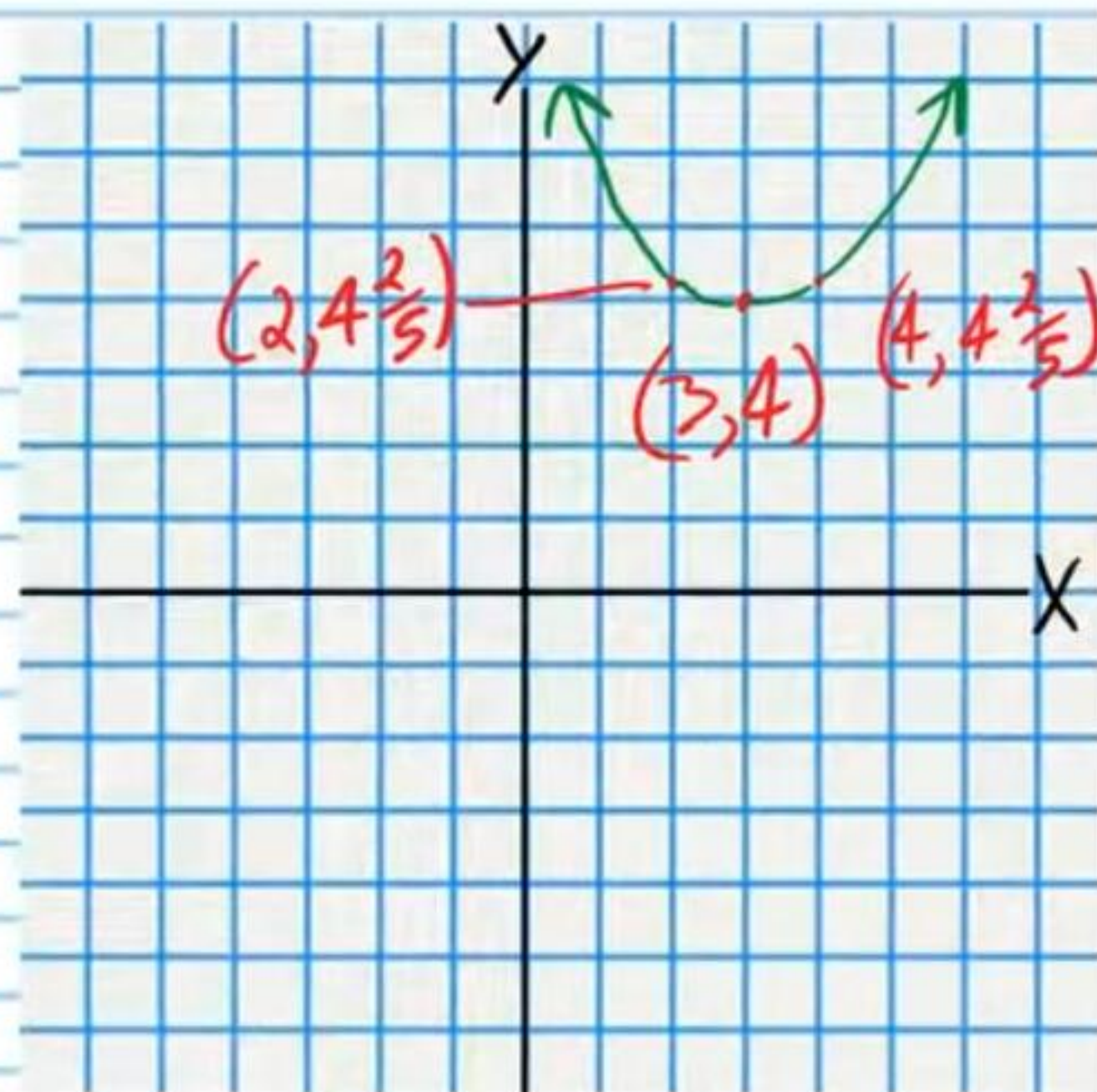
$$\text{II) } x = 2$$

$$y = \frac{2}{5}(2-3)^2 + 4$$

$$y = \frac{2}{5}(-1)^2 + 4$$

$$y = \frac{2}{5}(1) + 4$$

$$y = 4\frac{2}{5}$$



Graphing Variations of Quadratic Expressions

$$\textcircled{9} \quad y = 3x^2$$

$$y = 3(x-0)^2 + 0$$

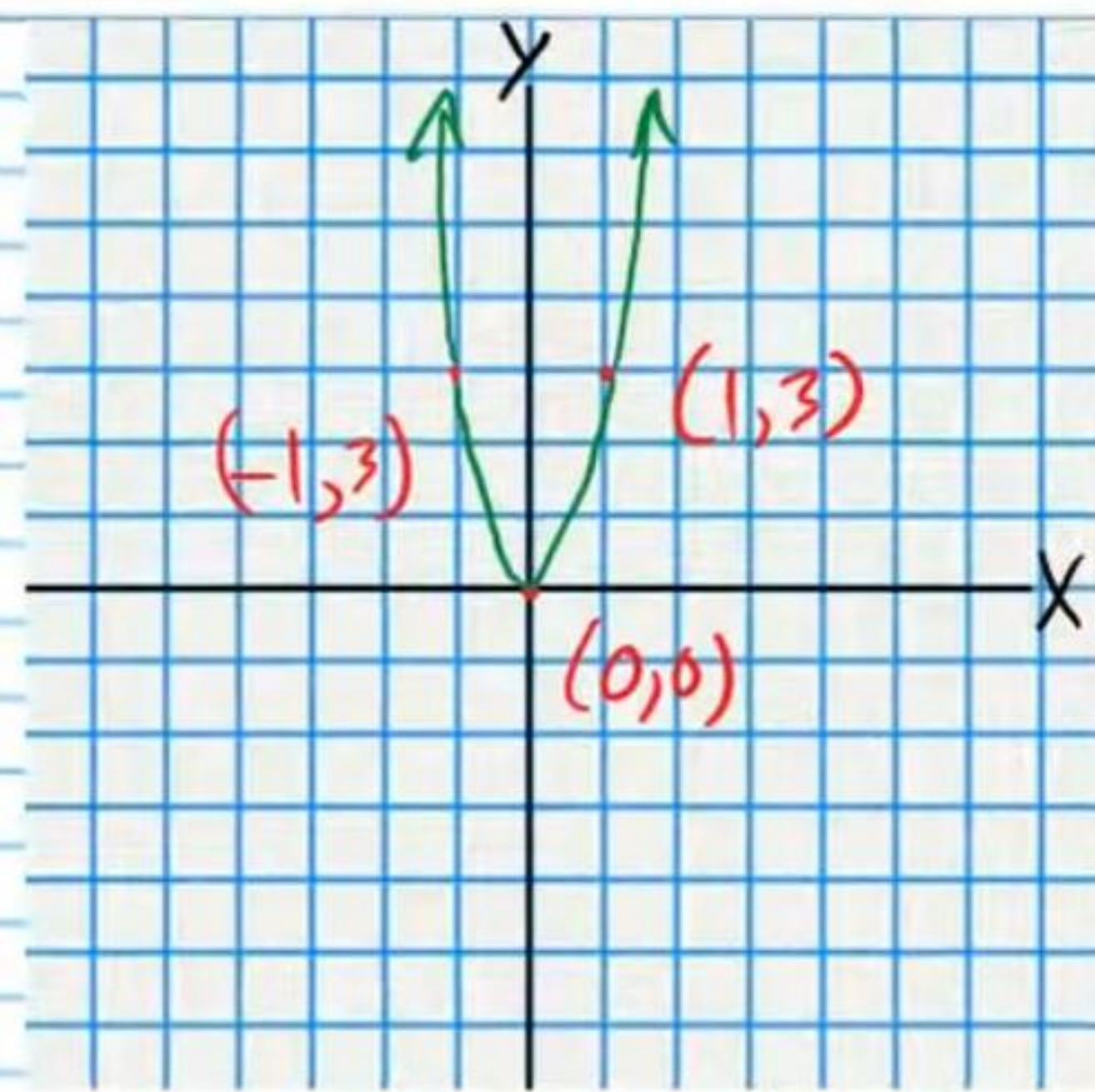
$$a = 3, h = 0, k = 0$$

$$\text{I) } (0, 0)$$

$$\text{II) } x = 1$$

$$y = 3(1)^2$$

$$y = 3$$



$$\textcircled{10} -\frac{x^2}{2} = y$$

$$y = -\frac{1}{2}x^2$$

$$y = -\frac{1}{2}(x-0)^2 + 0$$

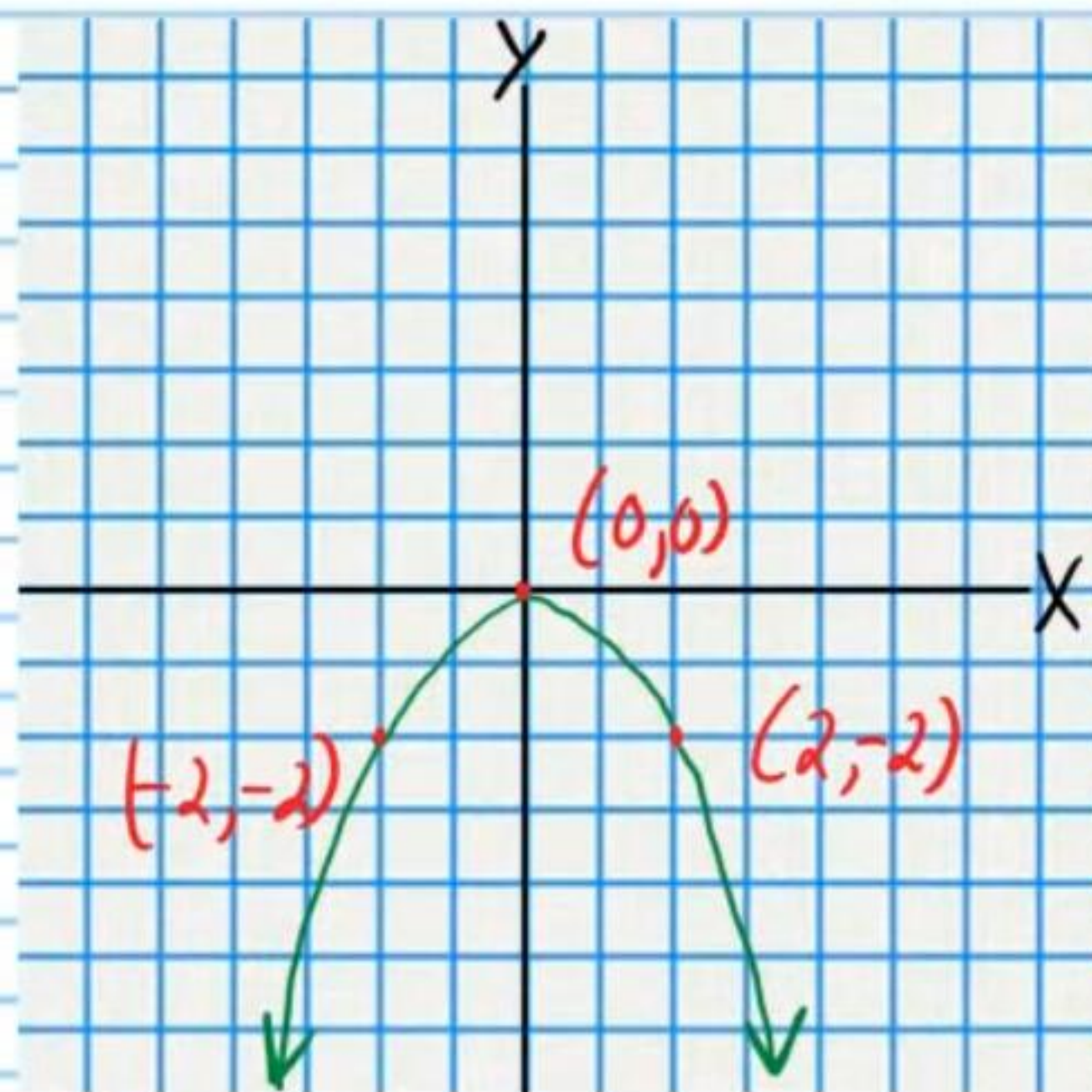
$$a = -\frac{1}{2}, h = 0, k = 0$$

$$\text{I) } (0,0)$$

$$\text{II) } x = 2$$

$$y = -\frac{(2)^2}{2}$$

$$y = -2$$



$$\textcircled{11} -x^2 = y$$

$$y = -(x-0)^2 + 0$$

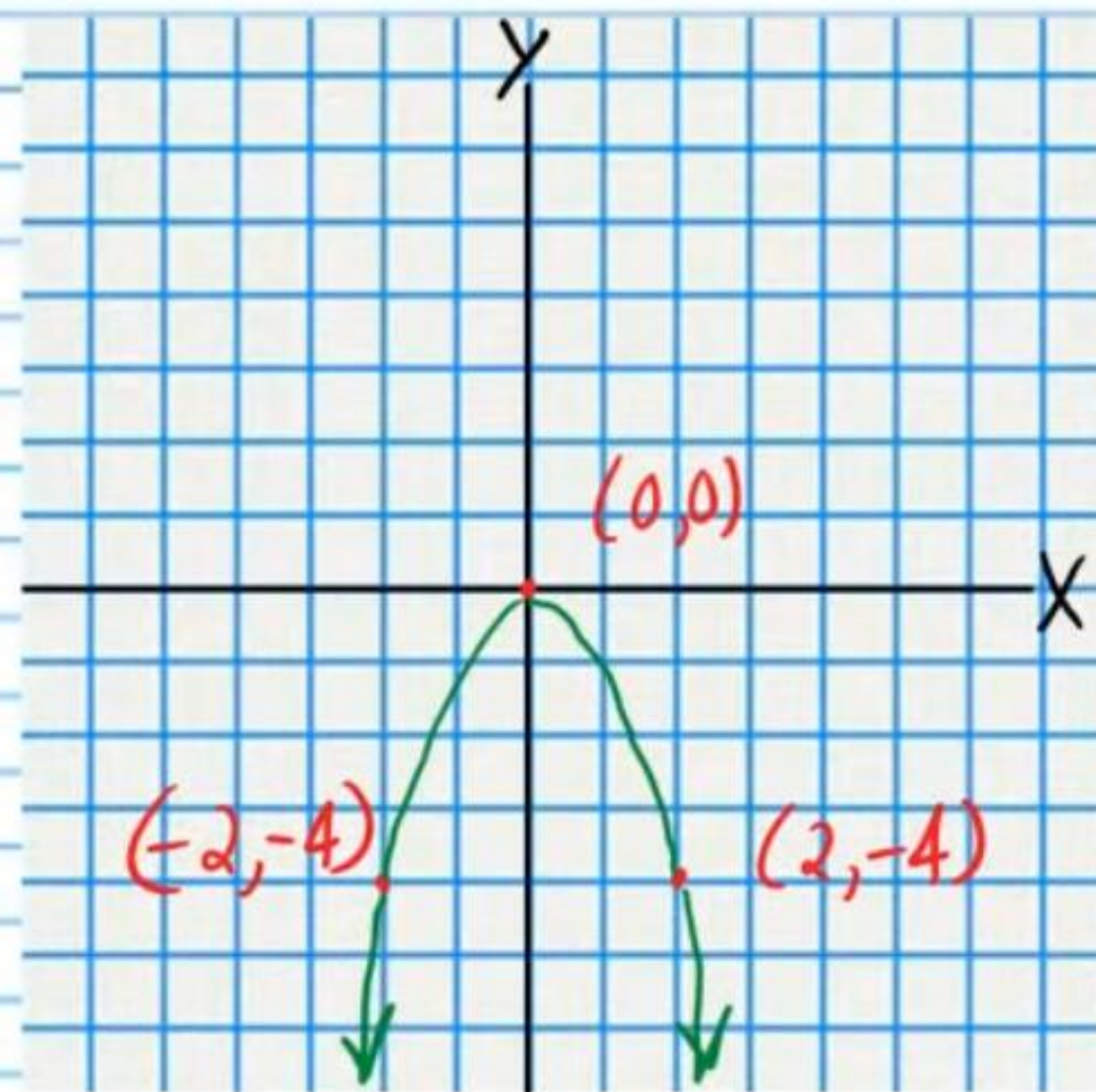
$$a = -1, h = 0, k = 0$$

$$\text{I) } (0,0)$$

$$\text{II) } x = 2$$

$$y = -(2)^2$$

$$y = -4$$



⑫ $y = \frac{3}{2}x^2$

$$y = \frac{3}{2}(x-0)^2 + 0$$

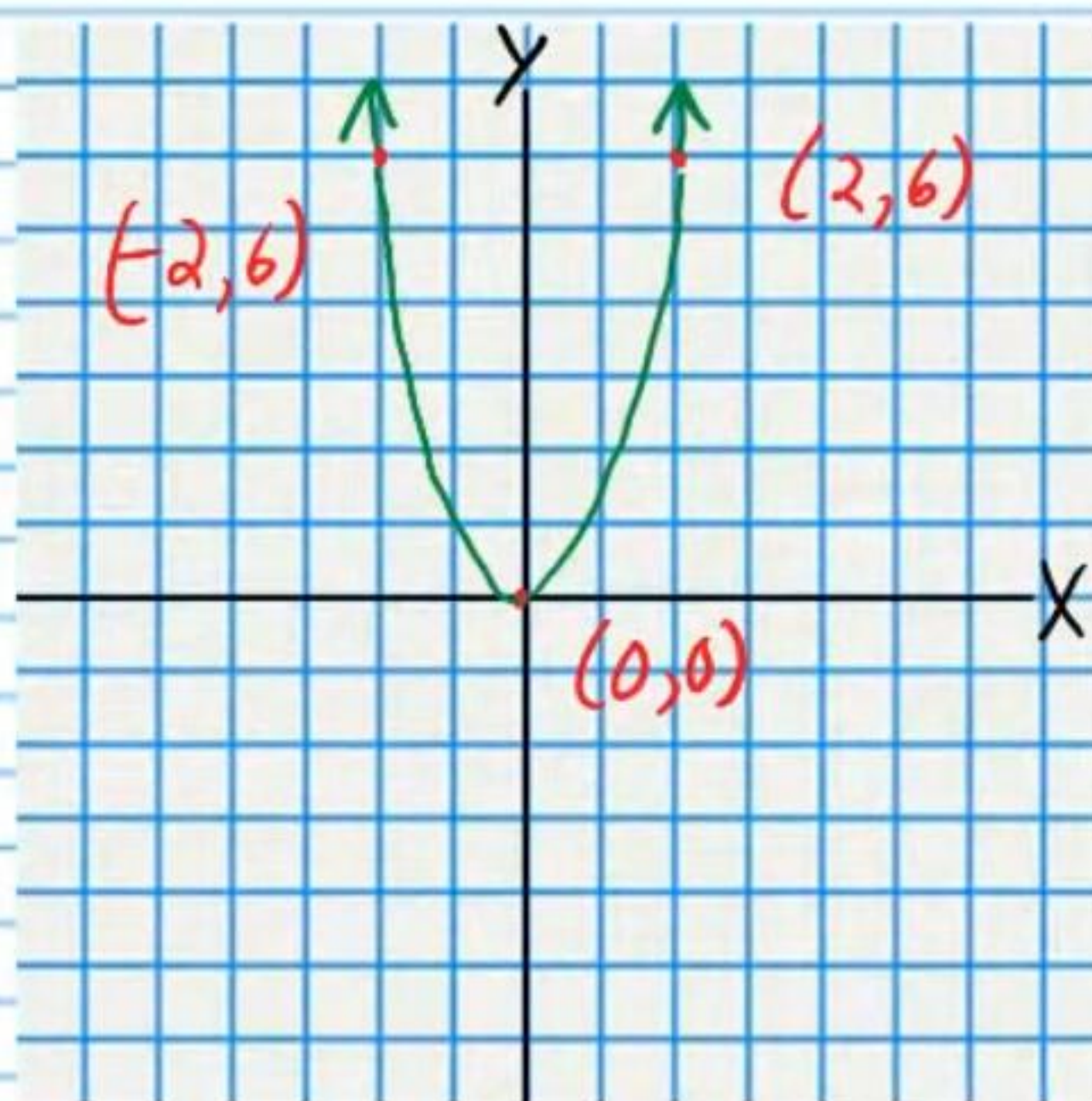
$$a = \frac{3}{2}, h = 0, k = 0$$

I) $(0,0)$

II) $x = 2$

$$y = \frac{3}{2}(2)^2$$

$$y = 6$$



⑬ $y = x^2 - 4x$

$$y = x(x-4)$$

$$y = (x-0)(x-4)$$

$$a = 1, m = 0, n = 4$$

I) $(0,0)$ & $(4,0)$

II) $V_x = \frac{m+n}{2}$

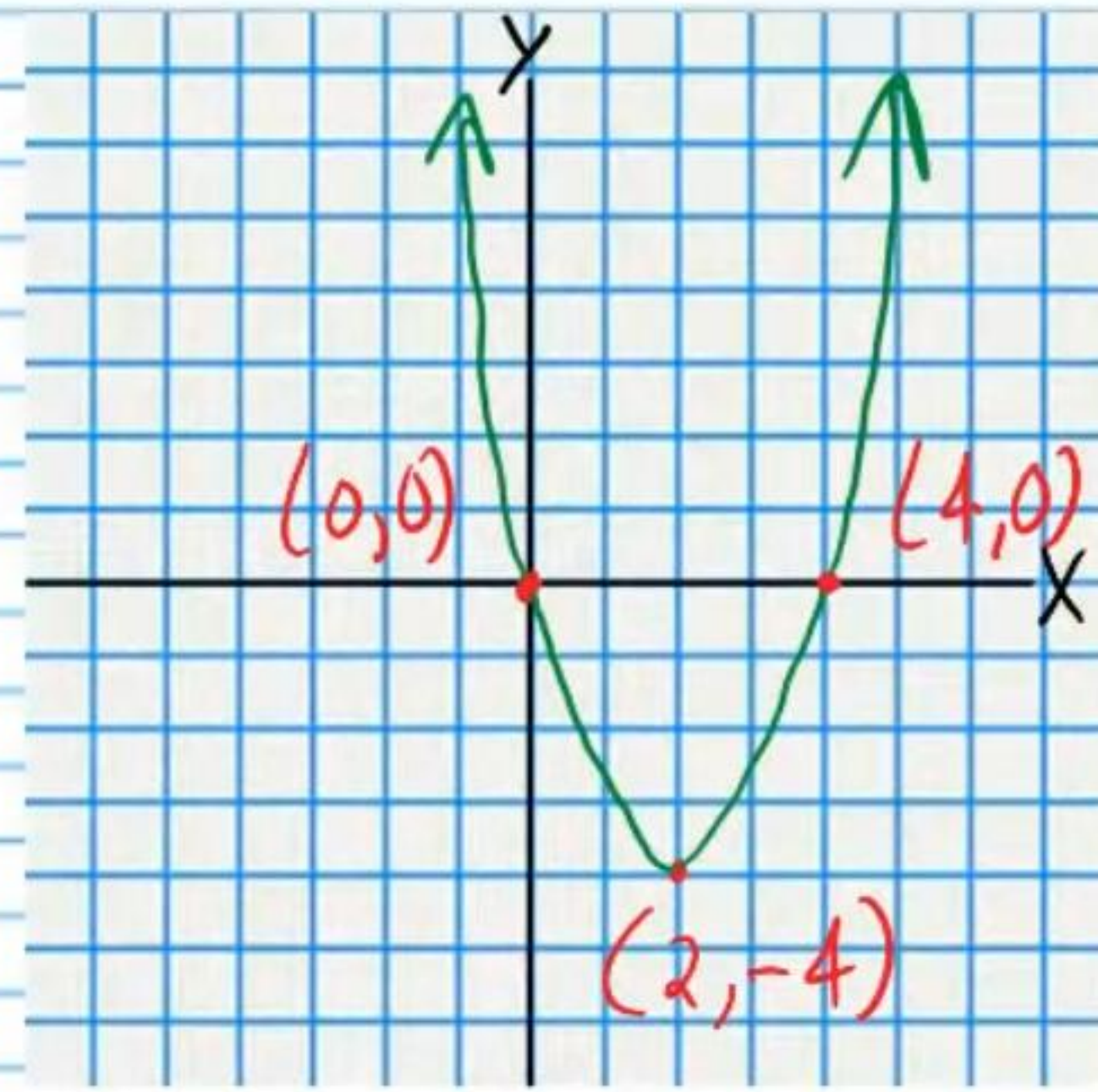
$$V_x = \frac{0+4}{2}$$

$$V_x = \frac{4}{2}$$

$$V_x = 2$$

$$V_y = 2(2-4)$$

$$V_y = -4$$



$$\textcircled{14} -3x^2 + 6x = y$$

$$y = -3x(x-2)$$

$$y = -3(x-0)(x-2)$$

$$a = -3, m = 0, n = 2$$

$$\text{I) } (0,0) \text{ \& } (2,0)$$

$$\text{II) } V_x = \frac{m+n}{2}$$

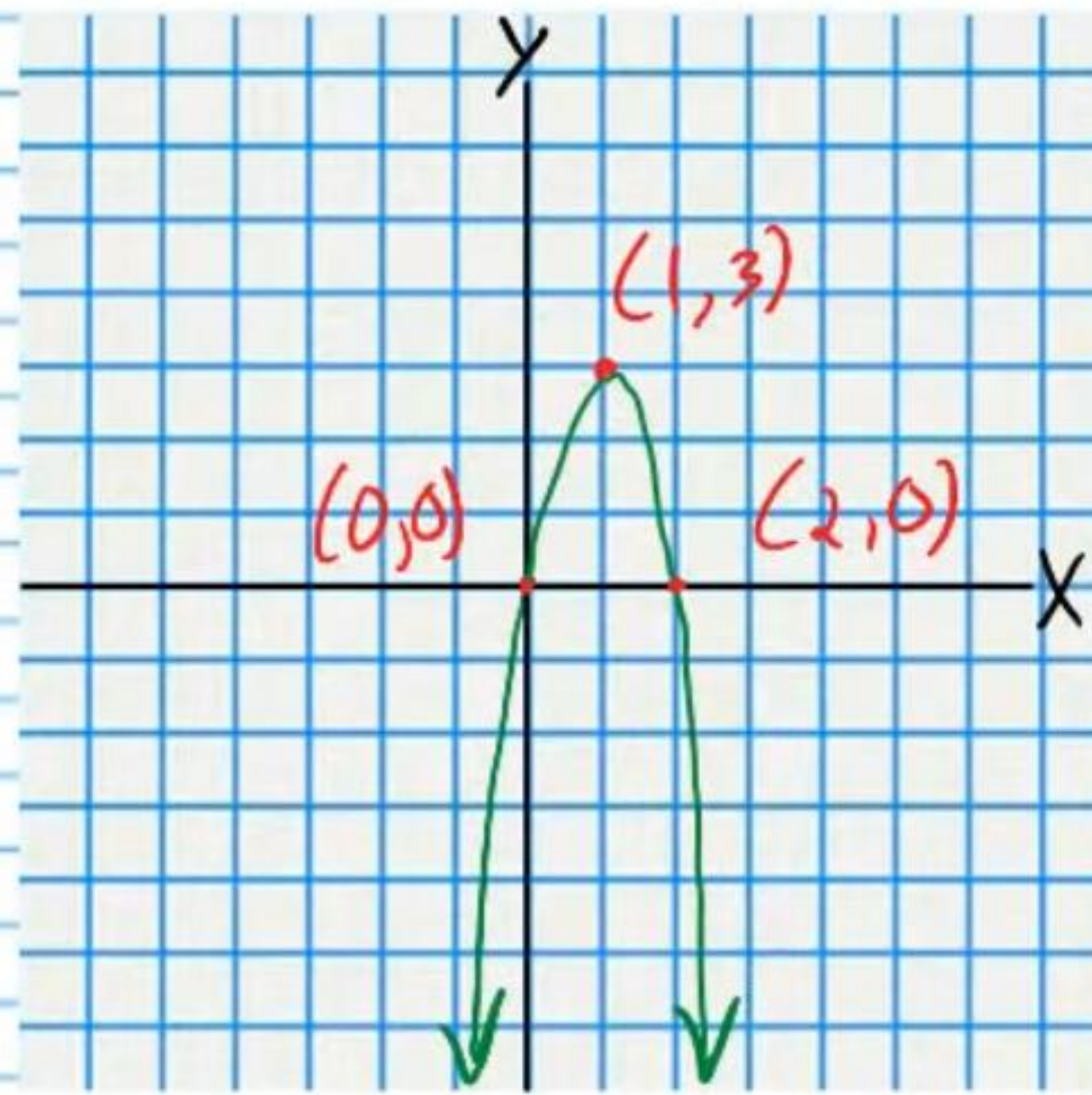
$$V_x = \frac{0+2}{2}$$

$$V_x = 1$$

$$V_y = -3(1)(1-2)$$

$$V_y = -3(1)(-1)$$

$$V_y = 3$$



$$\textcircled{15} x^2 - 6x = y$$

$$x(x-6) = y$$

$$(x-0)(x-6) = y$$

$$a = 1, m = 0, n = 6$$

$$\text{I) } (0,0) \text{ \& } (6,0)$$

$$\text{II) } V_x = \frac{m+n}{2}$$

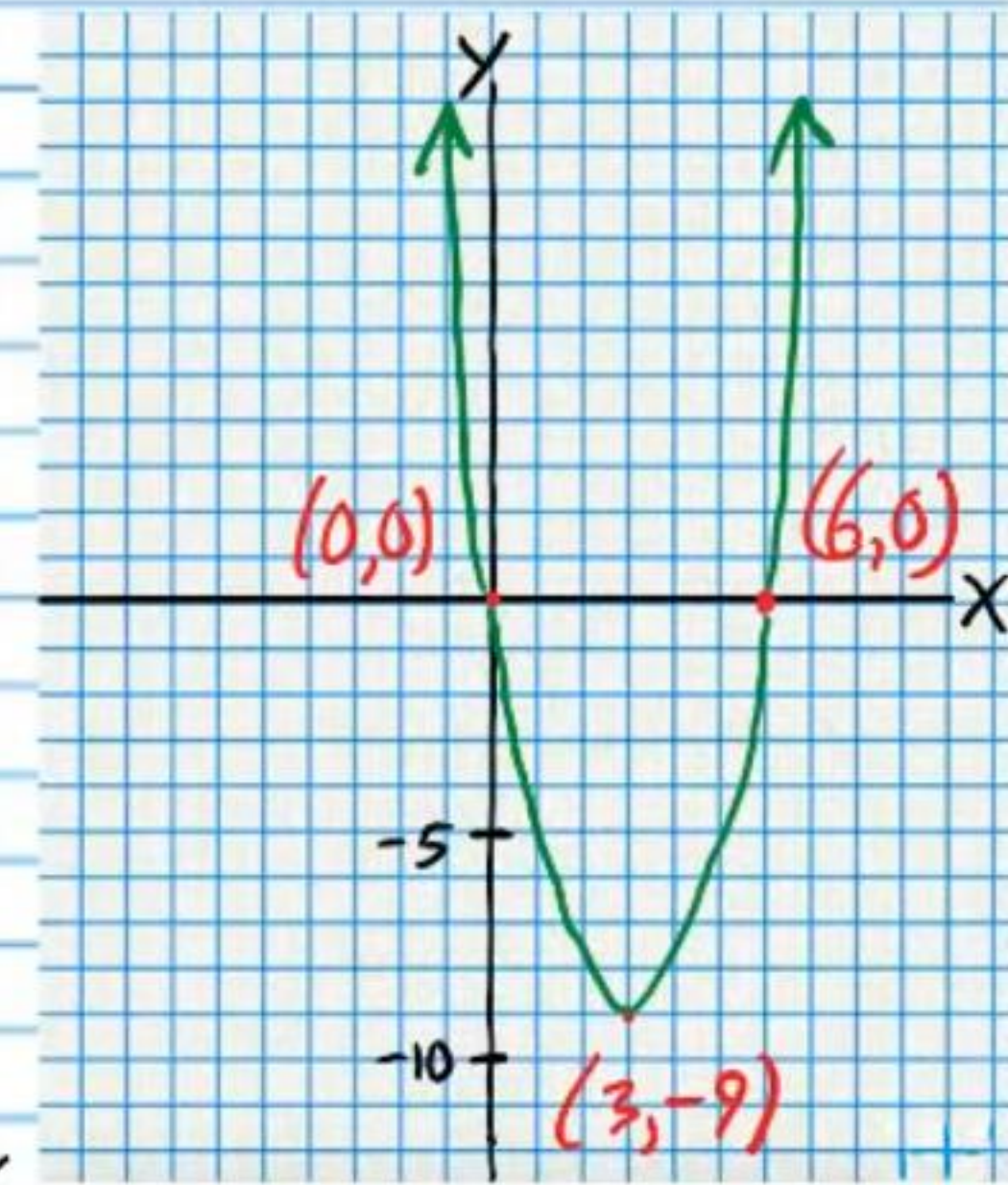
$$V_x = \frac{0+6}{2}$$

$$V_x = 3$$

$$V_y = 3(3-6)$$

$$V_y = 3(-3)$$

$$V_y = -9$$



* You may use a calculator.

$$\textcircled{16} y = -4x^2 - 12x$$

$$y = -4(x+3)$$

$$y = -4(x-0)(x+3)$$

$$a = -4, m = 0, n = -3$$

$$\text{I) } (0,0) \& (-3,0)$$

$$\text{II) } V_x = \frac{m+n}{2}$$

$$V_x = \frac{0+(-3)}{2}$$

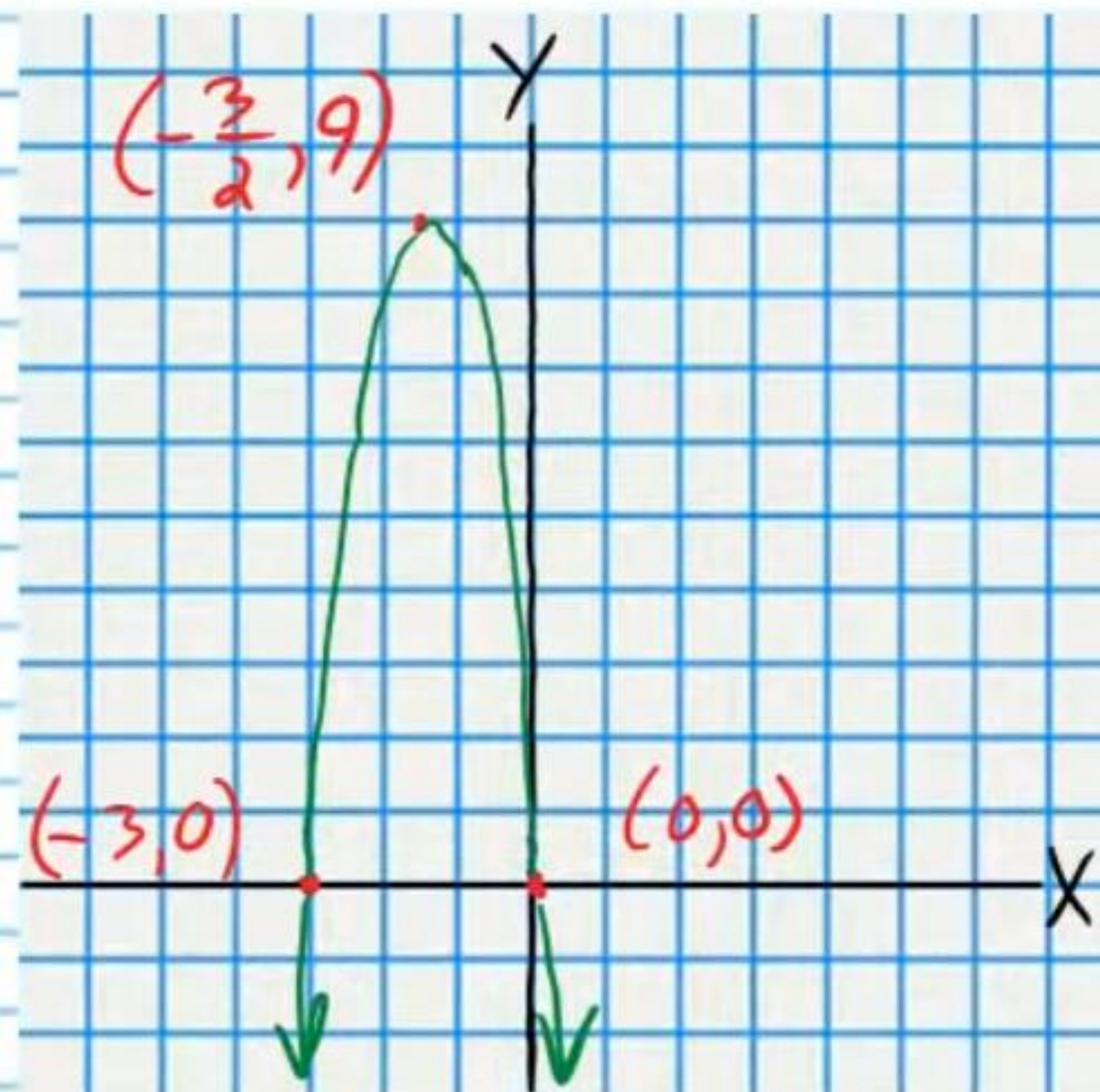
$$V_x = -\frac{3}{2}$$

$$V_y = -4\left(-\frac{3}{2}\right)\left(-\frac{3}{2}+3\right)$$

$$V_y = -4\left(-\frac{3}{2}\right)\left(\frac{3}{2}\right)$$

$$V_y = -4\left(-\frac{9}{4}\right)$$

$$V_y = 9$$



$$\textcircled{17} x(x+4) = y$$

$$(x-0)(x+4) = y$$

$$a = 1, m = 0, n = -4$$

$$\text{I) } (0,0) \& (-4,0)$$

$$\text{II) } V_x = \frac{m+n}{2}$$

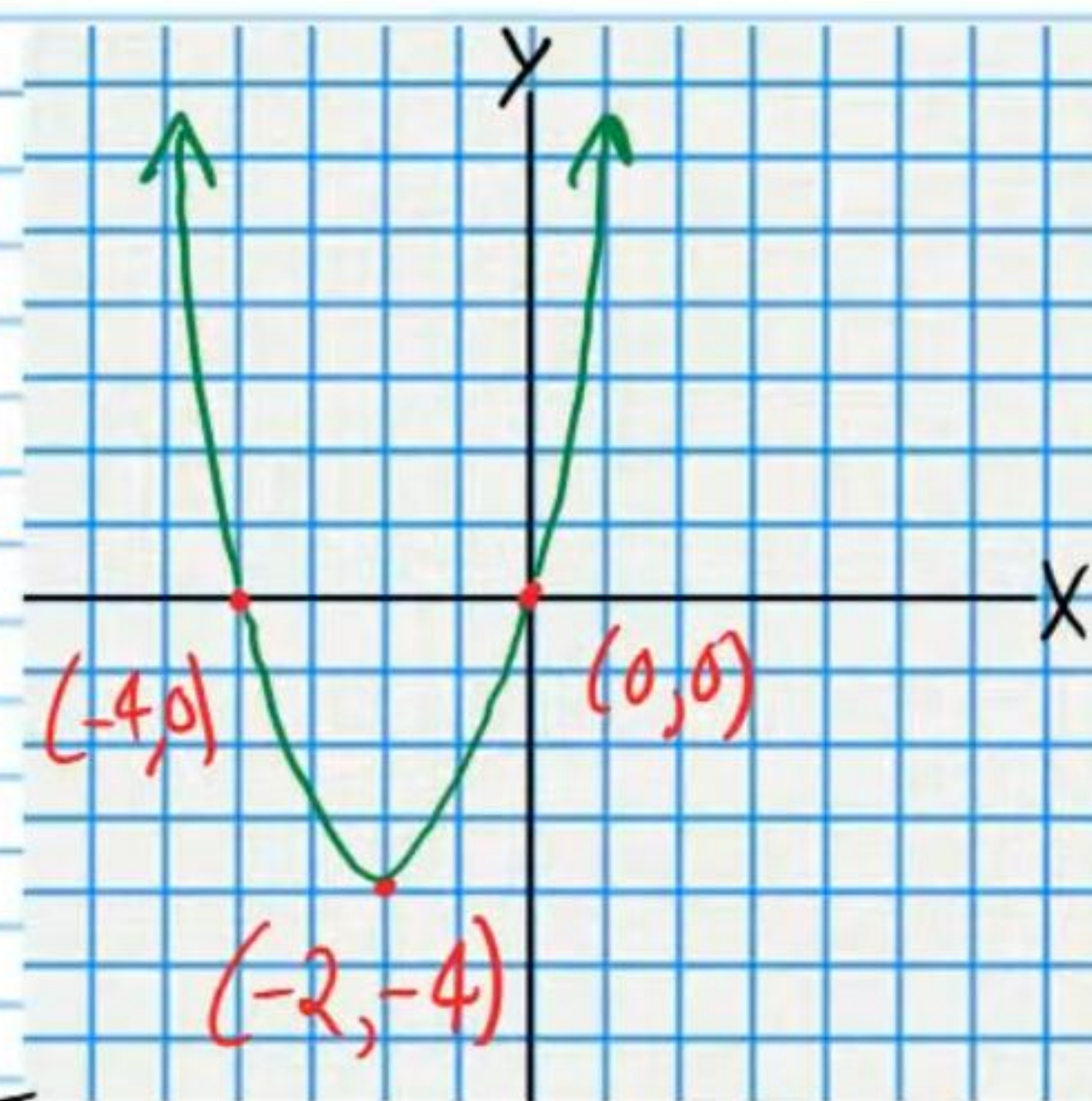
$$V_x = \frac{0+(-4)}{2}$$

$$V_x = -2$$

$$V_y = -2(-2+4)$$

$$V_y = -2(2)$$

$$V_y = -4$$



⑮ $y = 2x(x-6)$

$$y = 2(x-0)(x-6)$$

$$a=2, m=0, n=6$$

I) $(0,0)$ & $(6,0)$

II) $V_x = \frac{m+n}{2}$

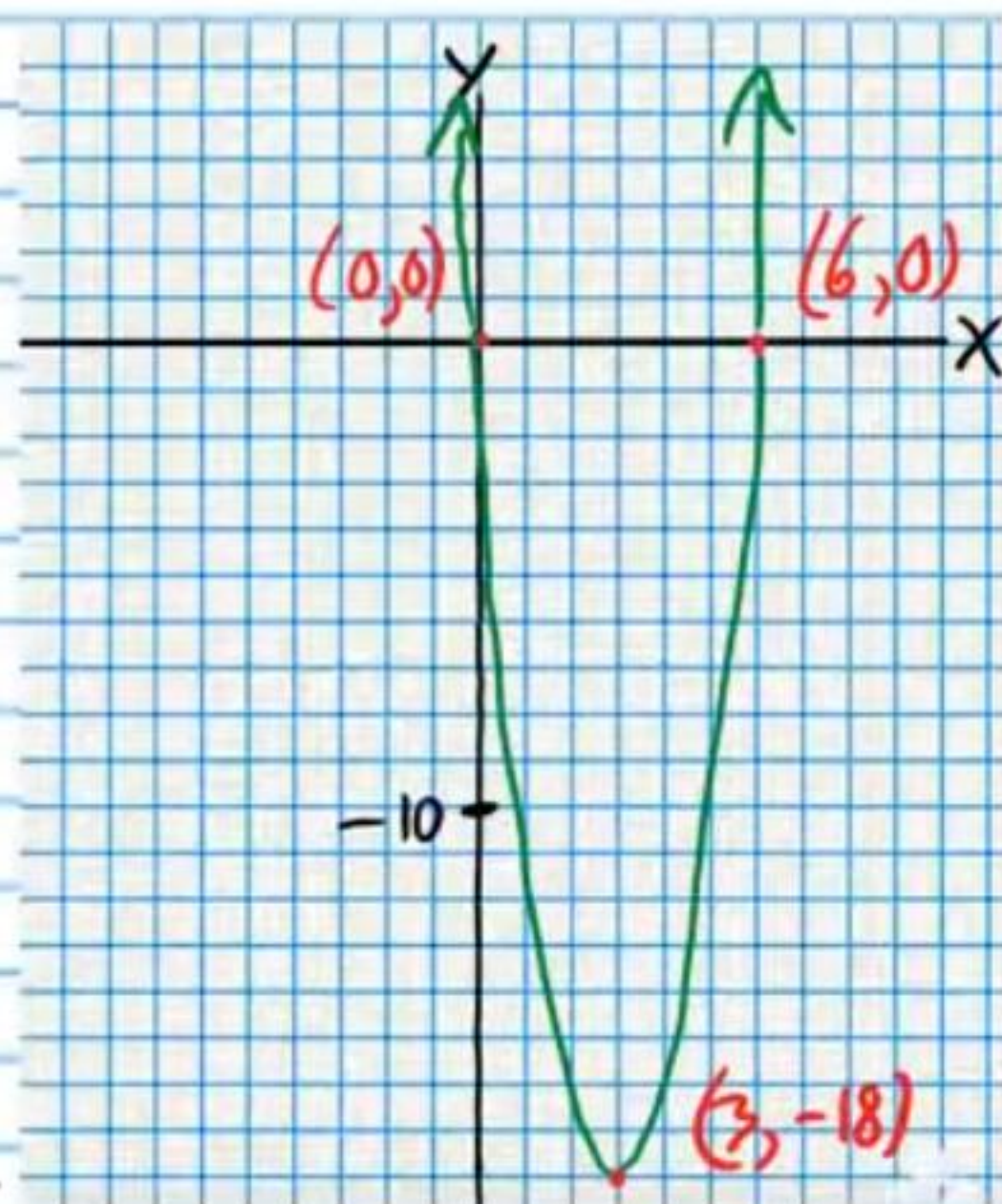
$$V_x = \frac{0+6}{2}$$

$$V_x = 3$$

$V_y = 2(3)(3-6)$

$$V_y = 6(-3)$$

$$V_y = -18$$



⑯ $y = 3x^2 - 2$

$$y = 3(x-0)^2 - 2$$

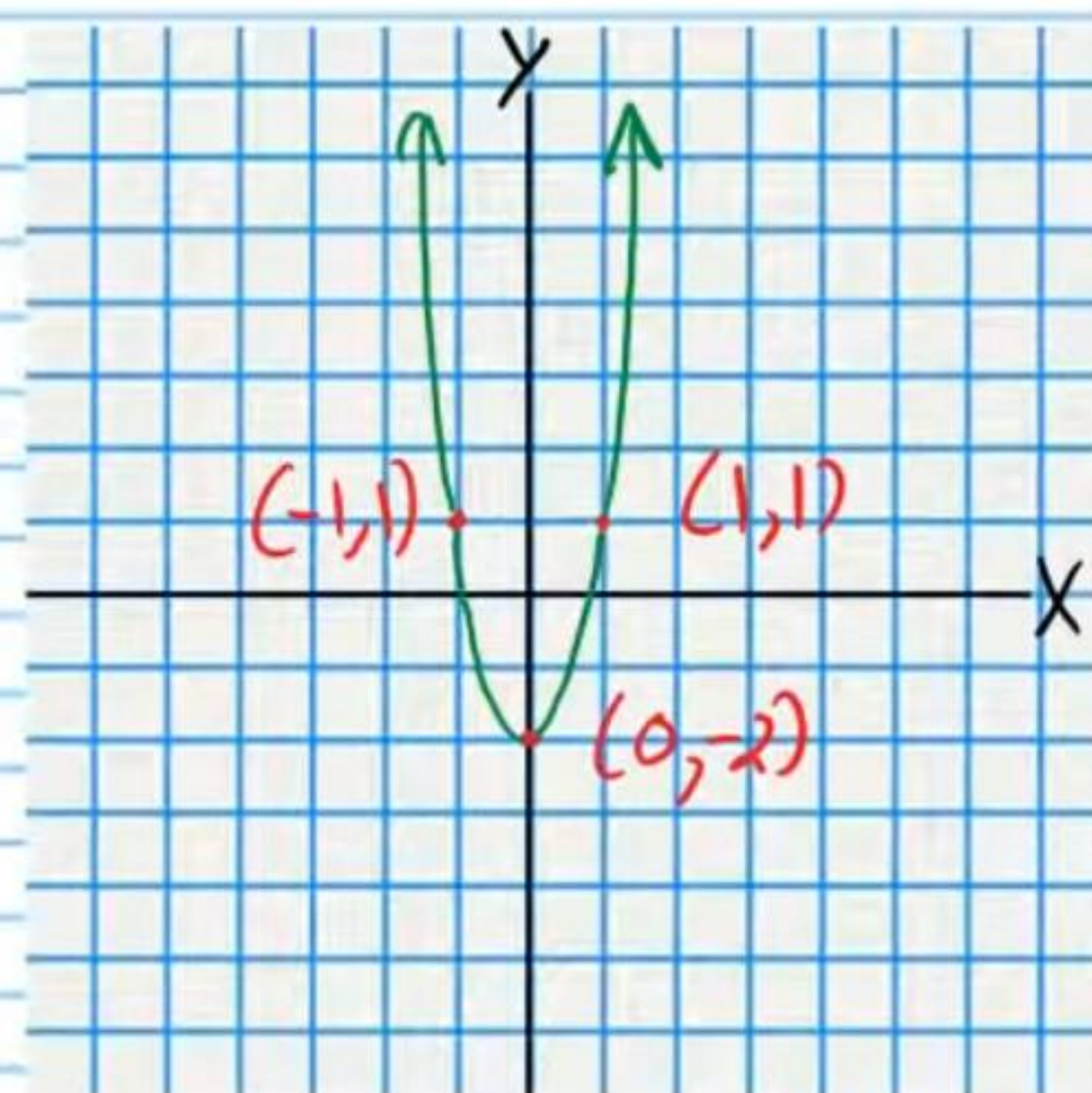
$$a=3, h=0, k=-2$$

I) $(0, -2)$

II) $x = 1$

$$y = 3(1)^2 - 2$$

$$y = 1$$



$$\textcircled{20} -2x^2 + 1 = y$$

$$-2(x-0)^2 + 1 = y$$

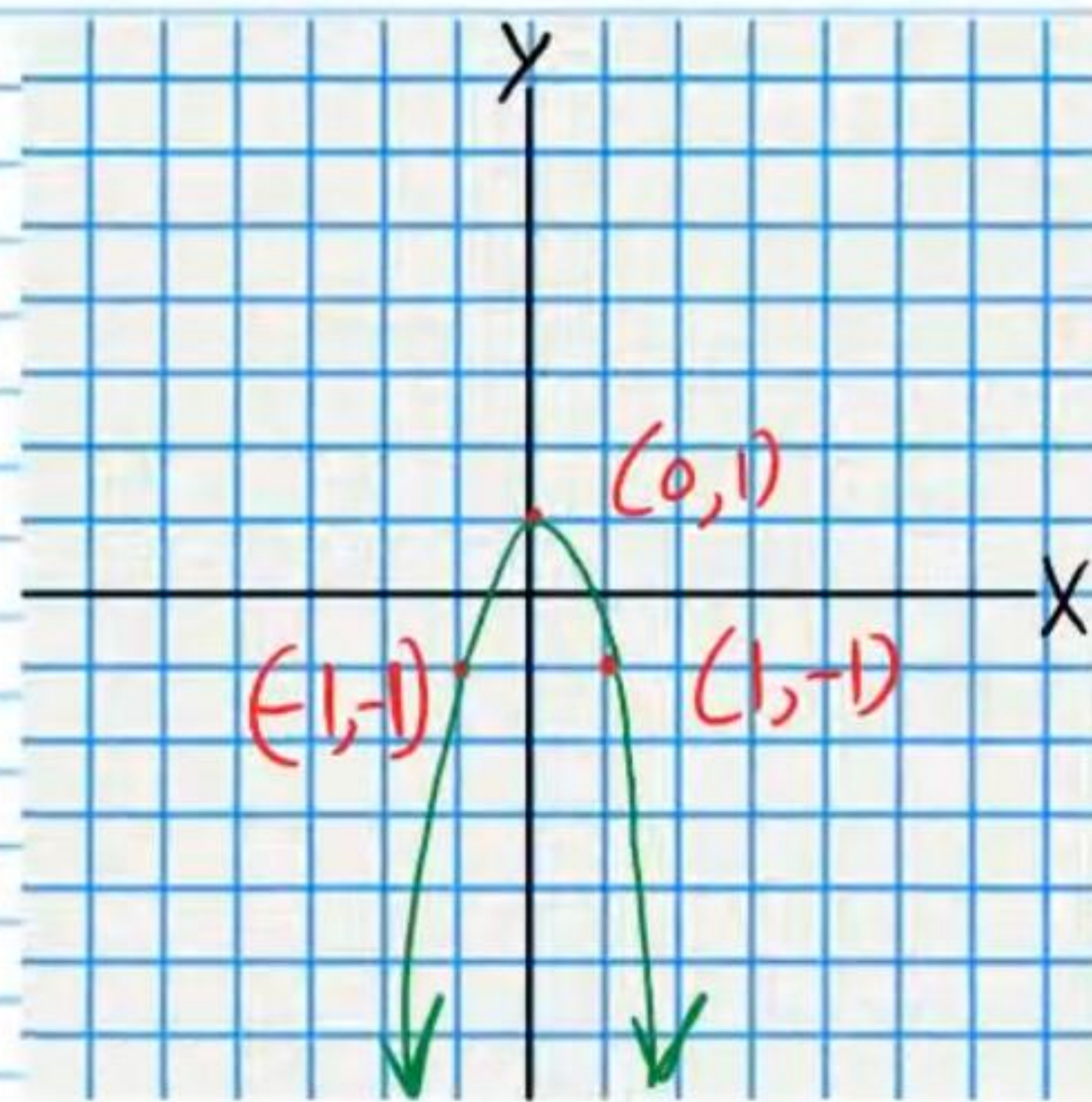
$$a = -2, h = 0, k = 1$$

$$\text{I) } (0, 1)$$

$$\text{II) } x = 1$$

$$y = -2(1)^2 + 1$$

$$y = -1$$



$$\textcircled{21} 2(x+3)^2 = y$$

$$2(x+3)^2 + 0 = y$$

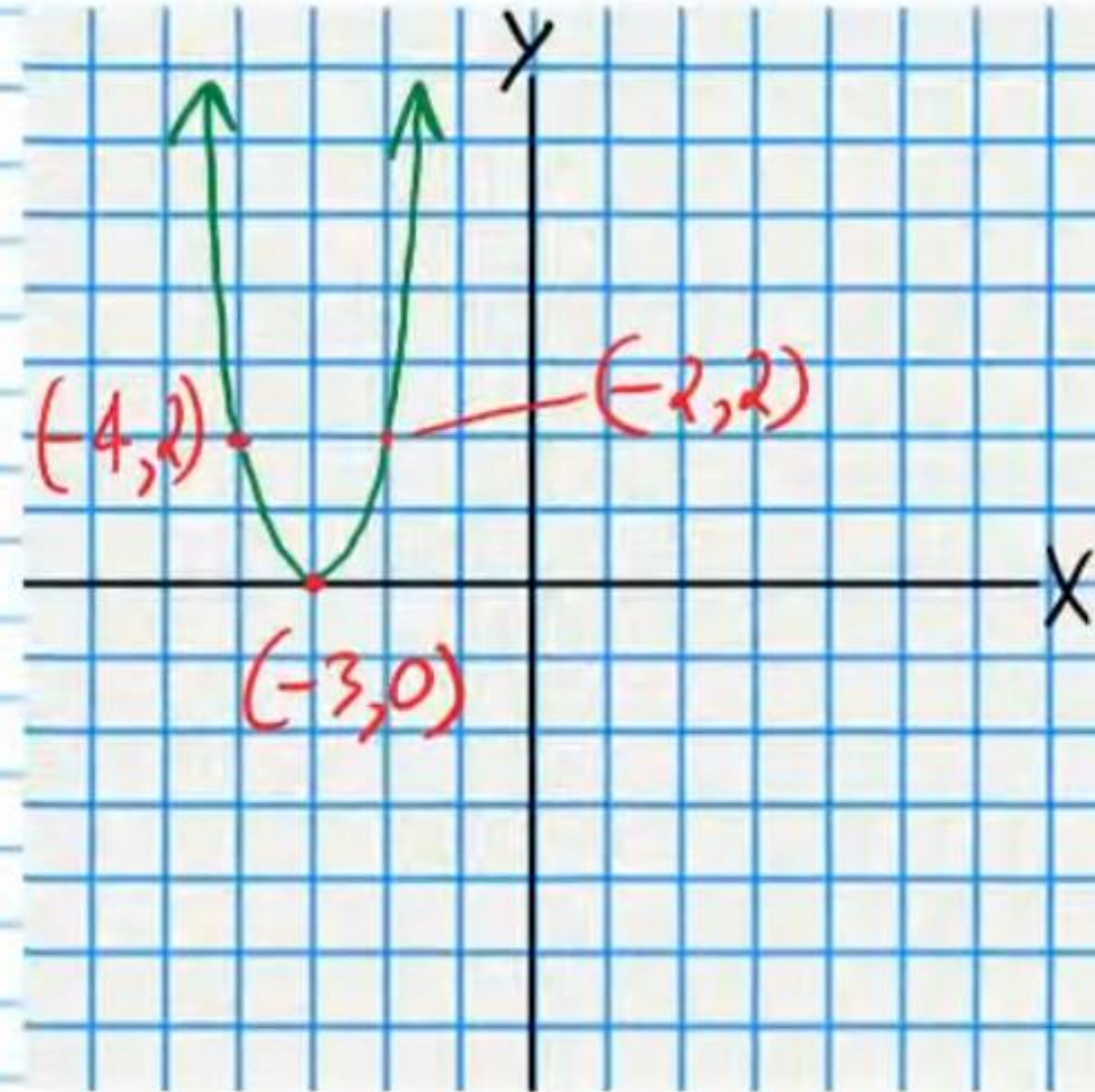
$$a = 2, h = -3, k = 0$$

$$\text{I) } (-3, 0)$$

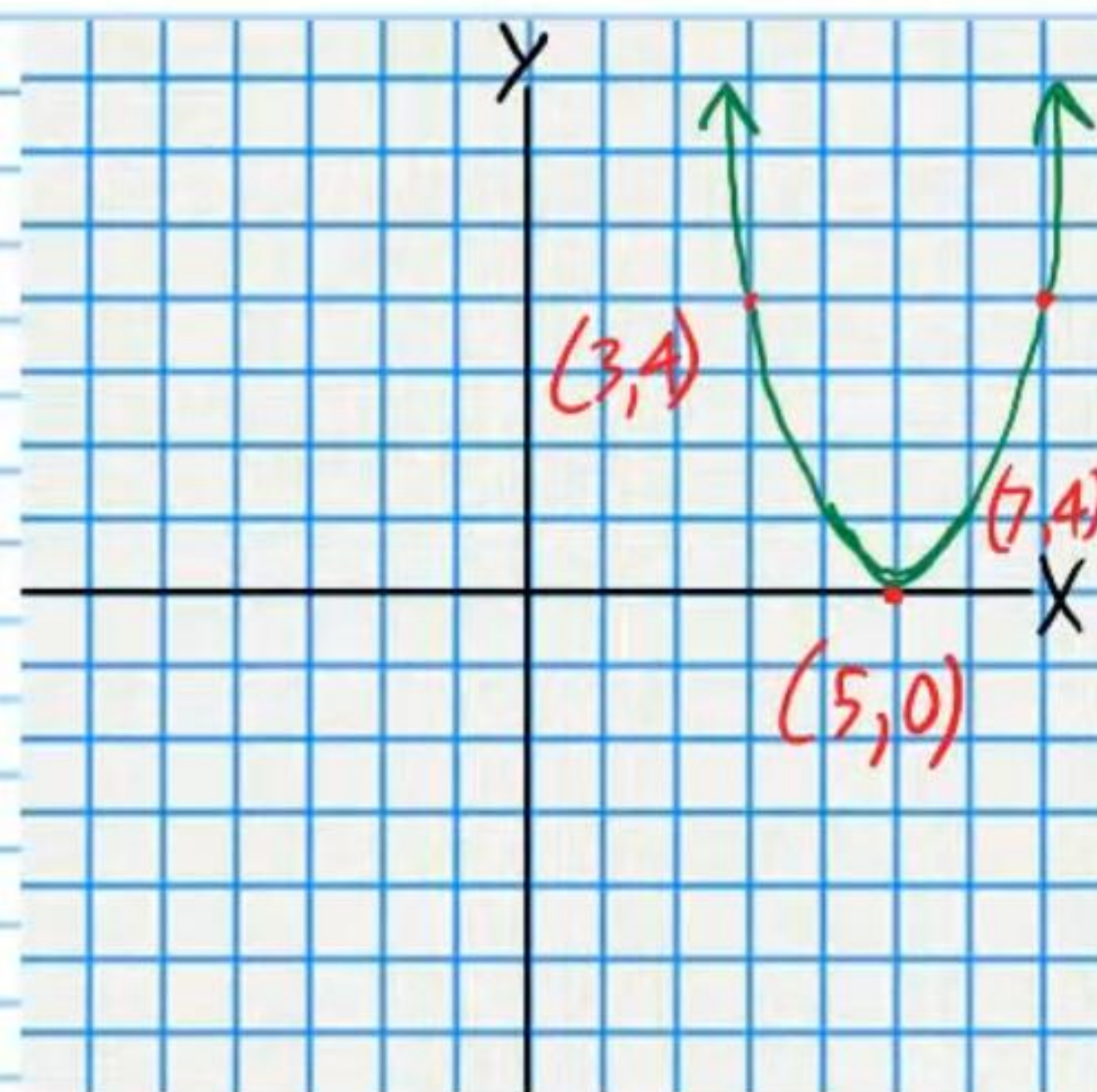
$$\text{II) } x = -2$$

$$y = 2(-2+3)^2 + 0$$

$$y = 2$$

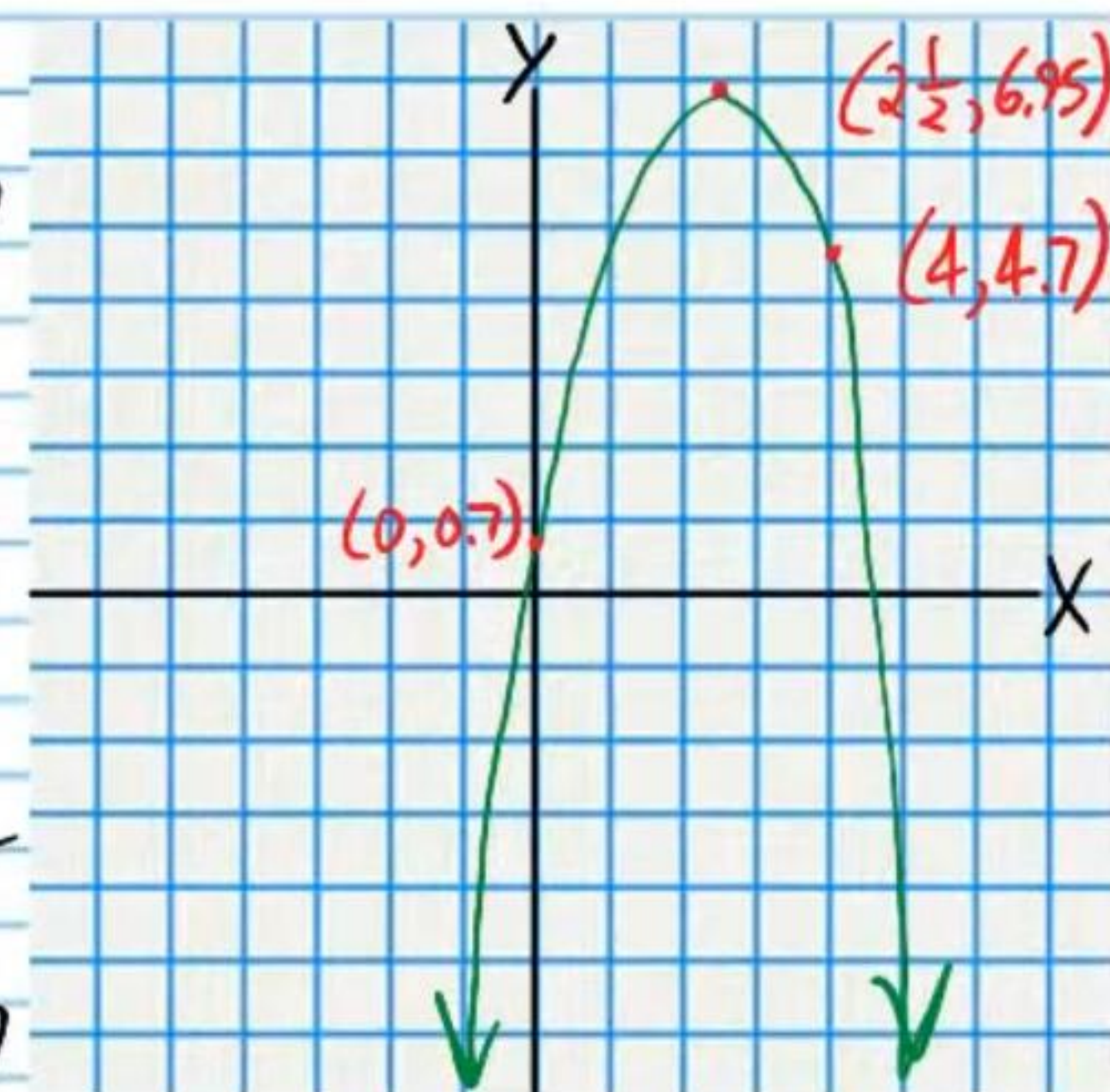


② $y = (x-5)^2$
 $y = (x-5)^2 + 0$
 $a=1, h=5, k=0$
 I) $(5,0)$
 II) $x=3$
 $y = (3-5)^2$
 $y = (-2)^2$
 $y = 4$



③ $y = -x^2 + 5x + 0.7$
 $a=-1, b=5, c=0.7$
 I) $V_x = -\frac{b}{2a}$
 $V_x = -\frac{5}{2(-1)}$
 $V_x = \frac{5}{2} = 2\frac{1}{2}$

$V_y = -(\frac{5}{2})^2 + 5(\frac{5}{2}) + 0.7$
 $V_y = -\frac{25}{4} + \frac{25}{2} + 0.7$
 $V_y = -\frac{25}{4} + \frac{50}{4} + 0.7$
 $V_y = \frac{25}{4} + 0.7$
 $V_y = 6\frac{1}{4} + 0.7$
 $V_y = 6.25 + 0.70$
 $V_y = 6.95$



II) $x=0$
 $y=0.7$

III) $x=4$
 $y = -(4)^2 + 5(4) + 0.7$
 $y = -16 + 20 + 0.7$
 $y = 4.7$

$$\textcircled{24} (x + \frac{1}{2})(x + 3) = y$$

$$a = 1, m = -\frac{1}{2}, n = -3$$

$$\text{I) } (-\frac{1}{2}, 0) \text{ \& } (-3, 0)$$

$$\text{II) } V_x = \frac{m+n}{2}$$

$$V_x = \frac{-\frac{1}{2} + (-3)}{2}$$

$$V_x = \frac{-\frac{7}{2}}{2}$$

$$V_x = -\frac{7}{2} \div 2$$

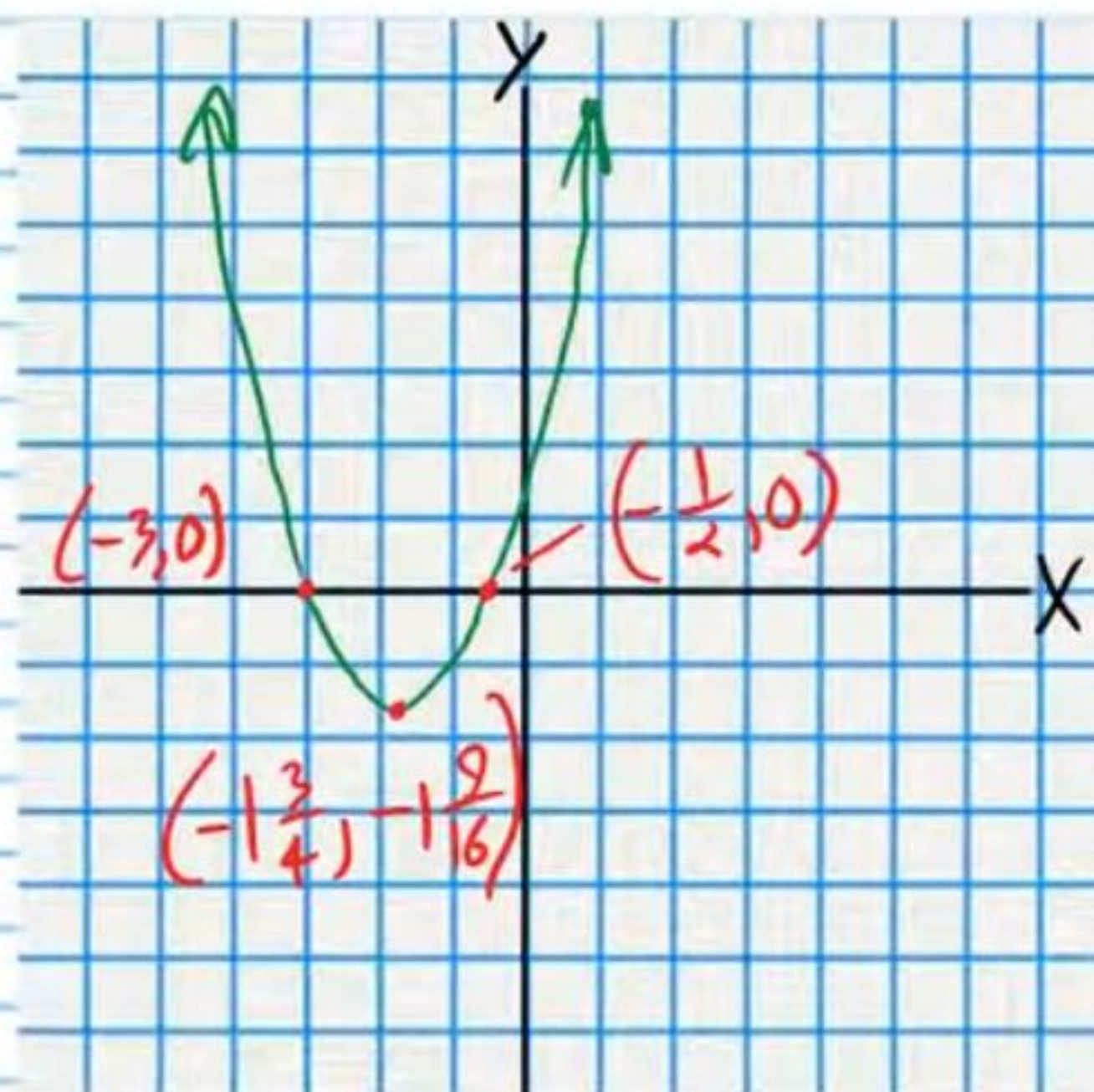
$$V_x = -\frac{7}{2} \cdot \frac{1}{2}$$

$$V_x = -\frac{7}{4} = -1\frac{3}{4}$$

$$\text{III) } V_y = (-\frac{7}{4} + \frac{1}{2})(-\frac{7}{4} + 3)$$

$$V_y = (-\frac{5}{4})(\frac{5}{4})$$

$$V_y = -\frac{25}{16} = -1\frac{9}{16}$$



$$\textcircled{25} y = -2(x + \frac{1}{3})^2 + 2.5$$

$$a = -2, h = -\frac{1}{3}, k = 2.5$$

$$\text{I) } (-\frac{1}{3}, 2.5)$$

$$\text{II) } x = \frac{2}{3}$$

$$y = -2(\frac{2}{3} + \frac{1}{3})^2 + 2.5$$

$$y = -2(1)^2 + 2.5$$

$$y = -2(1) + 2.5$$

$$y = -2 + 2.5$$

$$y = 0.5$$

$$\text{III) } x = -1\frac{1}{3}$$

$$y = -2(-1\frac{1}{3} + \frac{1}{3})^2 + 2.5$$

$$y = -2(-1)^2 + 2.5$$

$$y = -2(1) + 2.5$$

$$y = -2 + 2.5$$

$$y = 0.5$$

